

Nonlinear Programming (NLP)

Nonlinear programming (NLP) is the process of solving an optimization problem where some of the constraints or the objective function are nonlinear.

In the history of methodological development, the first attempts at dealing with inequality constraints were concentrated on linear ones only. With linearity prevailing in the constraints as well as in the objective function, linear programming (LP) is used to solve the problem. NLP improves the limitation of linearity by specifying the choice variables to be nonnegative, which is quite appropriate in most economic analysis. Besides, NLP makes it possible to handle nonlinear inequality constraints and nonlinear objective function. Thus NLP occupies a most important place in the optimization methodology.

Kuhn-Tucker Conditions:

In the classical optimization problem, with no explicit restrictions on the signs of the choice variables, and with no inequalities in the constraints, the first-order condition for a relative or local extremum is simply that the first partial derivatives of the Lagrangian function with respect to all the choice variables and the Lagrangian multipliers be zero. In NLP, there exists a similar type of first-order condition, known as the Kuhn-Tucker conditions. While the classical first-order condition is always necessary, the Kuhn-Tucker conditions can be the necessary conditions only when a certain proviso is satisfied. On the other hand, under certain

specific circumstances, the Kuhn-Tucker conditions turn out to be sufficient conditions, or even necessary-and-sufficient conditions as well.

The Kuhn-Tucker conditions can be developed in two steps:

Step 1: Effect of Nonnegativity Restrictions

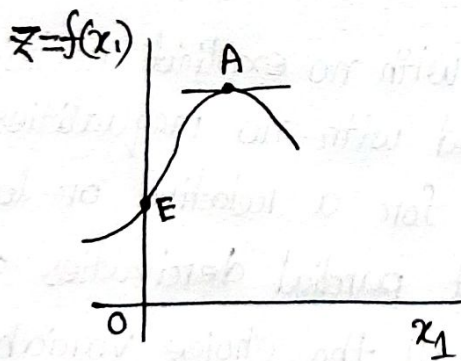
At first, consider a problem with nonnegativity restrictions on the choice variables, but with no other constraints. Taking the single-variable case, in particular, we have,

$$\text{Maximize } z = f(x_1) \quad \text{--- (1)}$$

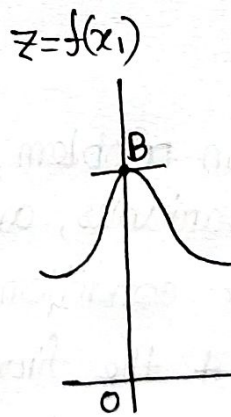
$$\text{Subject to } x_1 \geq 0$$

where the function f is assumed to be differentiable. In view of the restriction $x_1 \geq 0$, three possible situations may arise as shown in Figure 1.

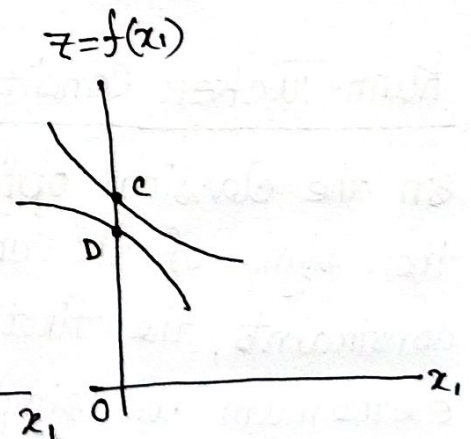
Figure 1.



(a)



(b)



(c)

First, if a local maximum of z occurs in the interior of the feasible region, as shown in point A of Figure 1(a), then we have an interior solution. The first-order condition in this case is $\frac{dz}{dx_1} = 0$; same as in the classical problem.

Second, as illustrated by point B in Fig. 1(b), a local maximum can occur on the vertical axis, where $x_1 = 0$. Therefore, there is a boundary solution where $\frac{dz}{dx_1} = 0$ remains valid.

Now, as a third possibility, a local maximum can take the position of point C or point D in Fig. 1(c), because these two points are higher than the neighboring point within the feasible region. Therefore, for the third possibility, we get $\frac{dz}{dx_1} < 0$.

By summing, it can be concluded that, in order for a value of x_1 to give a local maximum of z in problem (1), it must satisfy one of the following three conditions

$$\frac{dz}{dx_1} = 0 \quad \text{and} \quad x_1 > 0 \quad [\text{Point A}]$$

$$\frac{dz}{dx_1} = 0 \quad \text{and} \quad x_1 = 0 \quad [\text{Point B}]$$

$$\frac{dz}{dx_1} < 0 \quad \text{and} \quad x_1 = 0 \quad [\text{points C and D}]$$

These three conditions can be consolidated into a single statement

$$\frac{dz}{dx_1} \leq 0 \quad x_1 \geq 0 \quad \text{and} \quad \underset{\text{between}}{x_1 \frac{dz}{dx_1}} = 0 \quad \text{--- (2)}$$

The third part of (2) indicates that $\uparrow$ the two quantities x_1 and $\frac{dz}{dx_1}$, at least one must take a zero value, so that the product of these two must be zero. This feature is referred to as the complementary slackness between x_1 and $\frac{dz}{dx_1}$.

When the problem contains n choice variables:

$$\text{Maximize } z = f(x_1, x_2, x_3, \dots, x_n)$$

$$\text{Subject to } x_j \geq 0$$

— (3)

then the first-order condition can be written as:

$$\frac{\partial z}{\partial x_j} \leq 0 \quad x_j \geq 0 \quad \text{and} \quad x_j \frac{\partial z}{\partial x_j} = 0 \quad (j=1, 2, \dots, n)$$

— (4)

Step 2: Effect of Inequality Constraints

The second step include inequality constraints besides the nonnegativity constraints. Consider the following maximization problem:

$$\text{Maximize } \pi = f(x_j)$$

$$\text{Subject to } g^i(x_j) \leq \pi_i$$

$$\begin{pmatrix} j=1, 2, 3 \\ i=1, 2 \end{pmatrix}$$

— (5)

$$\text{and } x_j \geq 0$$

which, with the help of dummy variables s_i can be transformed into the equivalent form

$$\text{Maximize } \pi = f(x_j)$$

$$\text{subject to } g^i(x_j) + s_i = \pi_i$$

$$\begin{pmatrix} j=1, 2, 3 \\ i=1, 2 \end{pmatrix}$$

— (6)

$$\text{and } x_j, s_i \geq 0$$

If the nonnegativity restrictions are absent, we may, in line with the classical approach, form the Lagrangian function:

$$\mathcal{L} = f(x_j) + \lambda_i [\pi_i - g^i(x_j) - s_i] \quad \text{— (7)}$$

and write the first-order condition as

$$\frac{\partial \bar{z}'}{\partial x_j} = \frac{\partial \bar{z}'}{\partial \lambda_i} = \frac{\partial \bar{z}'}{\partial s_i} = 0$$

But since the x_j and s_i variables do have to be nonnegative, the first-order condition on those variables should be modified in accordance with (4). Therefore, we can write

$$\frac{\partial \bar{z}'}{\partial x_j} \leq 0 \quad x_j \geq 0 \quad \text{and} \quad x_j \frac{\partial \bar{z}'}{\partial x_j} = 0 \quad \text{--- (8)}$$

$$\frac{\partial \bar{z}'}{\partial s_i} \leq 0 \quad s_i \geq 0 \quad \text{and} \quad s_i \frac{\partial \bar{z}'}{\partial s_i} = 0$$

$$\frac{\partial \bar{z}'}{\partial \lambda_i} = 0 \quad \left[\begin{array}{l} \text{Since } g^i(x_j) + s_i = \pi_i; \text{ so } \pi_i - g^i(x_j) - s_i = 0 \\ \text{here } \frac{\partial \bar{z}'}{\partial \lambda_i} = \pi_i - g^i(x_j) - s_i = 0 \end{array} \right]$$

We can rewrite the (7) as

$$\bar{z} = f(x_j) + \lambda_i [\pi_i - g^i(x_j)] - \lambda_i s_i$$

$$\text{Now, } \frac{\partial \bar{z}'}{\partial x_j} = f_j - \lambda_i g_j^i$$

therefore, we can write the first line of (8) as

$$\frac{\partial \bar{z}'}{\partial x_j} = f_j - \lambda_i g_j^i \leq 0 \quad x_j \geq 0 \quad \text{and} \quad x_j \frac{\partial \bar{z}'}{\partial x_j} = 0$$

$$\text{Now, } \frac{\partial \bar{z}'}{\partial s_i} = -\lambda_i \leq 0 \quad \text{or, } \lambda_i \geq 0$$

therefore, we can write the second line of (8) as

$$\lambda_i \geq 0 \quad s_i \geq 0 \quad \text{and} \quad s_i \lambda_i = 0$$

$$\text{or, } s_i \geq 0 \quad \lambda_i \geq 0 \quad \text{and} \quad s_i \lambda_i = 0$$

$$\text{or, } \pi_i - g^i(x_j) \geq 0 \quad \lambda_i \geq 0 \quad \text{and} \quad \lambda_i [\pi_i - g^i(x_j)] = 0 \quad \left[\begin{array}{l} \text{Since } s_i = \pi_i - g^i(x_j) \\ \text{from equ. (6)} \end{array} \right]$$

Therefore, the Kuhn-Tucker condition of problem (5) can be stated as:

$$\frac{\partial z'}{\partial x_j} = f_j - \lambda_i g'_j \leq 0 \quad x_j \geq 0 \quad \text{and} \quad x_j \frac{\partial z'}{\partial x_j} = 0$$

$$r_i - g'_i(x_j) \geq 0 \quad \lambda_i \geq 0 \quad \text{and} \quad \lambda_i \frac{\partial z}{\partial \lambda_i} = 0$$

In general, the Kuhn-Tucker conditions for maximization problem are

$$\frac{\partial z}{\partial x_j} \leq 0 \quad x_j \geq 0 \quad \text{and} \quad x_j \frac{\partial z}{\partial x_j} = 0 \quad \left(\begin{matrix} i=1,2,\dots,m \\ j=1,2,\dots,n \end{matrix} \right)$$

$$\frac{\partial z}{\partial \lambda_i} \geq 0 \quad \lambda_i \geq 0 \quad \text{and} \quad \lambda_i \frac{\partial z}{\partial \lambda_i} = 0$$

Besides, the Kuhn-Tucker conditions for minimization are:

$$\frac{\partial z}{\partial x_j} \geq 0 \quad x_j \geq 0 \quad \text{and} \quad x_j \frac{\partial z}{\partial x_j} = 0$$

$$\frac{\partial z}{\partial \lambda_i} \leq 0 \quad \lambda_i \geq 0 \quad \text{and} \quad \lambda_i \frac{\partial z}{\partial \lambda_i} = 0 \quad \left(\begin{matrix} i=1,2,\dots,m \\ j=1,2,\dots,n \end{matrix} \right)$$

$$\text{Maximize } z = 2y - x^2$$

$$\text{Subject to } x^2 + y^2 \leq 1$$

$$\text{and } x, y \geq 0$$

The Lagrangian function is

$$z = 2y - x^2 + \lambda [1 - x^2 - y^2]$$

The Kuhn-Tucker conditions are

$$\frac{\partial z}{\partial x} = -2x - 2x\lambda \leq 0 \quad \bar{x} \geq 0 \quad \text{and} \quad \bar{x} \frac{\partial z}{\partial x} = 0$$

$$\frac{\partial z}{\partial y} = 2 - 2y\lambda \leq 0 \quad \bar{y} \geq 0 \quad \text{and} \quad \bar{y} \frac{\partial z}{\partial y} = 0 \quad \text{--- (1)}$$

$$\frac{\partial z}{\partial \lambda} = 1 - x^2 - y^2 \geq 0 \quad \bar{\lambda} \geq 0 \quad \text{and} \quad \bar{\lambda} \frac{\partial z}{\partial \lambda} = 0$$

To solve a nonlinear programming problem, the typical approach is one of trial and error.

Step 1: First check the possibility that $\bar{x} > 0$, $\bar{y} > 0$, but $\bar{\lambda} = 0$.

From complementary slackness we can write

$$-2x - 2x\lambda = 0 \quad [\text{since } \bar{x} > 0] \quad \text{and} \quad 2 - 2y\lambda = 0 \quad [\because \bar{y} > 0]$$

$$0 \pi, -2x(1+\lambda) = 0$$

$$0 \pi, -2x(1+0) = 0 \quad [\because \bar{\lambda} = 0]$$

$$0 \pi, -2x = 0$$

$$\therefore x = 0$$

$$0 \pi, 2y\lambda = 2$$

$$0 \pi, y\lambda = 1$$

$$0 \pi, y = \frac{1}{\lambda}$$

$$0 \pi, y = \frac{1}{0}$$

$$\therefore y = \infty$$

which violates the conditions.

Step 2: Check whether $\bar{x}$ or $\bar{y}$ equal to 0, but $\bar{\lambda} > 0$

if $\bar{x} = 0, \bar{y} = 1$ [$\because \bar{x}^2 + \bar{y}^2 \leq 1$]

Since $\bar{y} > 0$, therefore, $\frac{\partial Z}{\partial y} = 2 - 2y\lambda = 0$

but, hence $\frac{\partial Z}{\partial y} = 2 - 2 \cdot 1 \cdot \lambda(7,0) \neq 0$

if $\bar{y} = 0; \bar{x} = 1$

then $\frac{\partial Z}{\partial x}$ has to be 0.

but, hence $\frac{\partial Z}{\partial x} = -2x - 2\lambda x$

$= -2(1) - 2(\lambda) \cdot 1$

$= -2 - 2\lambda(7,0) \neq 0$

Therefore, these solutions violate the condition.

Step 3: Now check the KTC when $\bar{x} > 0, \bar{y} > 0$, and $\bar{\lambda} > 0$.

from complementary slackness

$-2x - 2\lambda x = 0$ ——— (2)

$2 - 2y\lambda = 0$ ——— (3)

$1 - \bar{x}^2 - \bar{y}^2 = 0$ ——— (4)

From (2) and (3) we can write

$-2x - 2\lambda x = 0$

and $2 - 2y\lambda = 0$

or, $-2x(1+\lambda) = 0$

or, $2 = 2y\lambda$

or, $\bar{x} = 0$

or, $\bar{y} = \frac{1}{\lambda}$

Now, substituting the value of x and y into (4) we get

$$1 - 0 - \frac{1}{\lambda^2} = 0$$

$$\text{or, } 1 = \frac{1}{\lambda^2}$$

$$\text{or, } \lambda^2 = 1$$

$$\therefore \lambda = 1$$

$$\text{So, } \bar{y} = \frac{1}{\lambda} = 1$$

Thus all the variables satisfy the nonnegativity constraints, then
the optimal solution $\bar{x} = 0$

$$\bar{y} = 1$$

$$\bar{\lambda} = 1$$

$$\text{and } \bar{z} = 2$$